NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Marketing Insights: Customer Segmentation and Predictive Modeling at NEW SMI

2026

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Marketing Engineering and Analytics Report

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I. Abstract

New Super Markets International (NEW SMI) is a Portuguese supermarket chain looking to move beyond mass marketing and make better use of its customer data. Using transactional and demographic information from 20K customers collected throughout 2025, this project develops a customer segmentation and a predictive model to support more focused marketing decisions. The segmentation identifies distinct customer groups based on purchasing behavior, enabling differentiated marketing actions for each segment. The predictive model targets a new home delivery subscription service, identifying the customers most likely to subscribe and maximizing campaign efficiency. All analyses were conducted in KNIME.

II. Introduction & Methodology

With over fifty years in the Portuguese retail sector, NEW SMI has accumulated a substantial customer database but has relied almost exclusively on undifferentiated marketing. As data-driven approaches become standard in retail, the company recognized the need to extract actionable insight from its data and develop more personalized customer strategies.

The dataset provided covers 20K randomly selected customers and includes variables on demographics, purchasing behavior, spending patterns, online activity, and satisfaction. The project was structured around two main objectives: customer segmentation and predictive modelling for a new delivery subscription service.

The work followed a sequential methodology. The dataset was first explored through descriptive analysis to identify inconsistencies, missing values, and outliers. Data preprocessing was then applied, covering data cleaning, variable transformation, and the creation of derived variables. Customer segmentation was subsequently performed to group customers with similar profiles and support differentiated marketing actions. Finally, predictive models were built and evaluated to identify customers with the highest subscription propensity, enabling a targeted and cost-efficient campaign. All stages were developed in KNIME.

1. Data Exploration and Understanding

1.1 Variable Description and Attributes of the Dataset

The dataset provided by NEW SMI is a customer-signature table comprising 20K randomly selected customers, with transactional data covering the full year of 2025 (a complete year was used to eliminate seasonality effects). Each observation represents a unique customer, identified by a CustID (substituted from the original RowID, as both variables carried the same meaning).

The dataset contains 16 variables, structured across four analytical dimensions: demographic profile, capturing customer characteristics such as age, income, and education; purchasing behavior (RFM), covering recency, frequency, and monetary value; product categories, reflecting spending across the five lines of business; and channel and satisfaction, measuring online engagement and customer sentiment. The full variable description, data types, and measurement levels are presented in Table 1.

Table 1 - Variable Description and Attributes table

Variable	Dimension	Description	Type	Measurement
CustID	-	Unique customer ID	String	Nominal
Education	Demographic	Academic Degree	String	Ordinal
Marital_Status	Demographic	Marital status	String	Nominal
Age	Demographic	Age in years	Integer	Continuous
Gender	Demographic	Gender	String	Nominal
Dependents	Demographic	If the customer has dependents in the household	Boolean	Binary
Income	Demographic	Household net income (yearly)	Double	Continuous
Frequency	RFM	Number of purchases (invoices) in 2025	Integer	Discrete
Monetary	RFM	Amount spent in 2025	Double	Continuous
Recency	RFM	# days since last purchase	Integer	
Perishables	Product Categories	Monetary per respective line of business (LoB) in 2025	Double	Continuous
Beverages	Product Categories		Double	Continuous
Frozen	Product Categories		Double	Continuous
Canned	Product Categories		Double	Continuous
Others	Product Categories		Double	Continuous
Internet	Channel & Satisfaction	% of sales over the Internet	Double	Continuous
NPS	Channel & Satisfaction	Adapted Net Promoter Score (1-5)	Integer	Ordinal

1.2 Exploratory Data Analysis

Before proceeding to data preparation, an exploratory statistical analysis was conducted to better understand the structure, distribution, and potential quality issues present in the dataset. This step informed all subsequent decisions regarding outlier treatment, missing value imputation, and feature engineering.

1.2.1 Exploratory Data Analysis of Key Variables

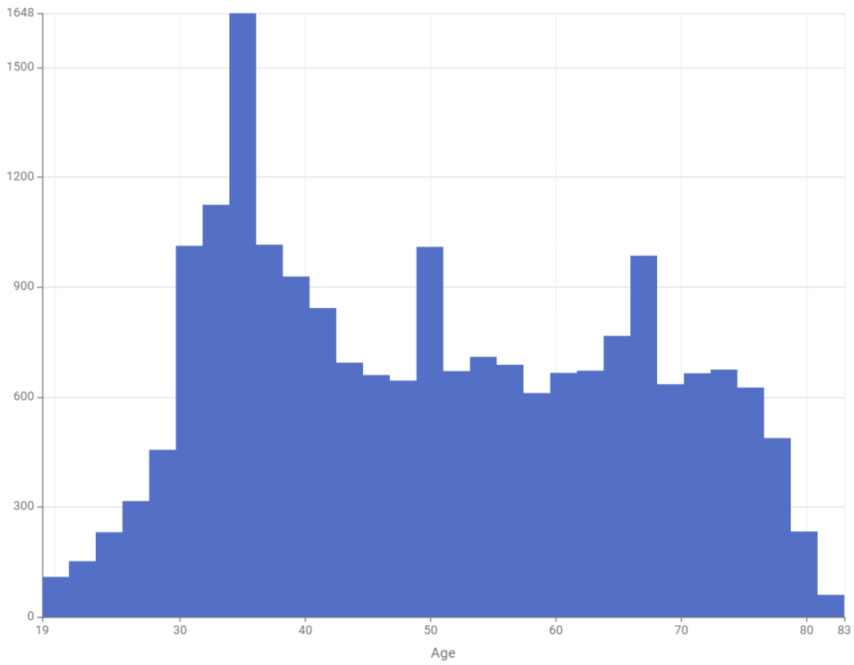

Figure 1 - Age Distribution of NEW SMI Customers

The age distribution of NEW SMI customers spans from 19 to 83 years, reflecting a broad and diverse customer base. The histogram reveals a multi-peaked distribution rather than a simple bell curve, with the most prominent concentration occurring among customers in their early to mid-thirties. A second notable peak is observed among customers aged approximately 60 to 70 years, suggesting that two age groups play a particularly significant role in the customer base: younger working-age adults and older senior customers. Customers below 30 years of age are present but less represented, which may point to an opportunity for NEW SMI to strengthen its appeal and outreach toward younger audiences. Overall, the distribution reinforces the relevance of age as a segmentation variable, given the clear behavioral and economic differences that tend to accompany different life stages.

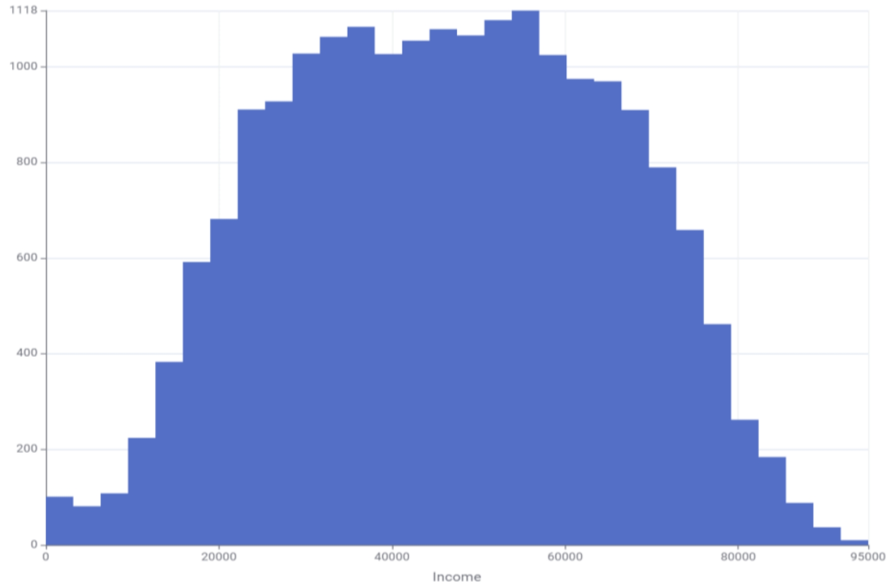

Figure 2 - Income Distribution of NEW SMI Customers

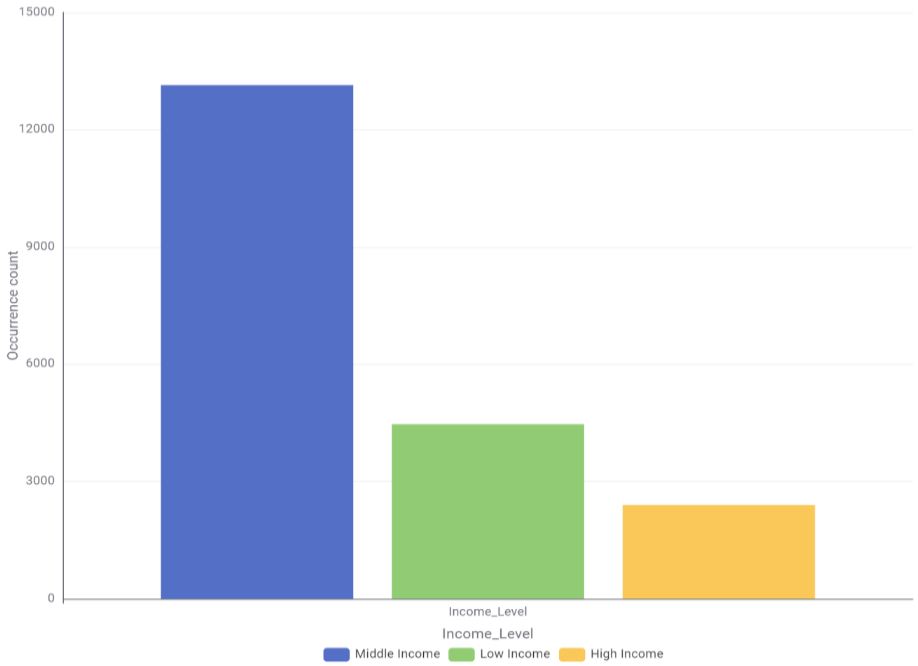

Figure 3 - Customer Distribution by Income Level

The income distribution of NEW SMI customers ranges from approximately €0 to €95,000 following the capping treatment applied during data preparation. The histogram reveals a broadly symmetrical distribution, which is somewhat unusual for income data, with the majority of customers concentrated between €30,000 and €70,000 and the highest density observed around the middle-income range. The estimated median income falls between €45,000 and €50,000, making it a more representative central tendency measure than the mean for this variable. This is further supported by the income-level segmentation, which shows that approximately 66% of customers belong to the middle-income group, around 22% to the low-income group and roughly 12% to the high-income group, confirming that NEW SMI primarily serves middle-income households. The smaller high-income segment, while limited in size, may nonetheless support selective premium positioning in specific product categories or channels.

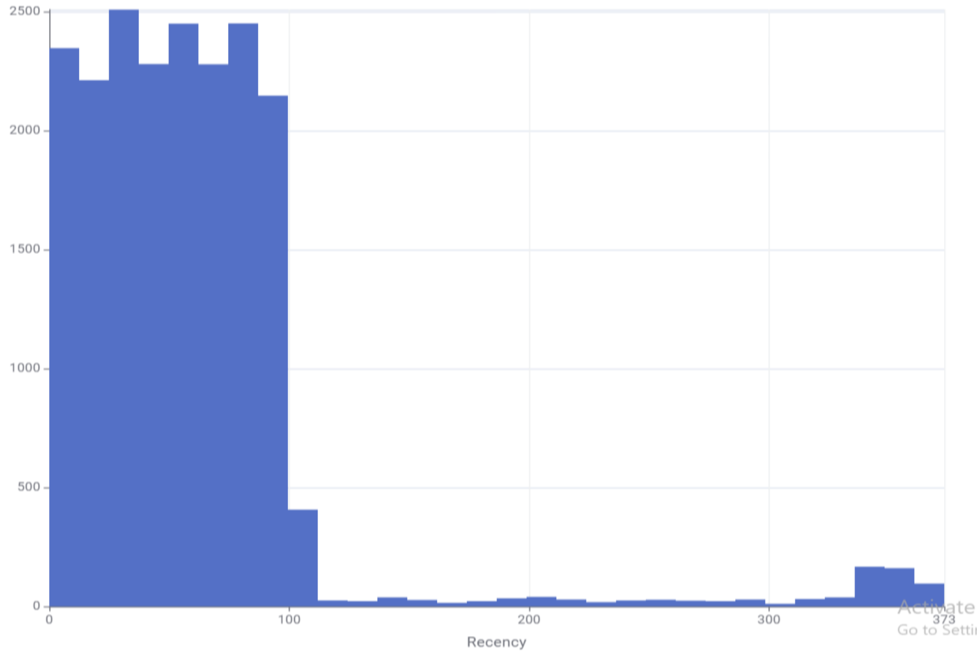

Figure 4 - Recency Distribution of NEW SMI Customers

The recency distribution ranges from 0 to 373 days, reflecting a wide spectrum of customer engagement levels. The histogram reveals a strongly right-skewed distribution, with the large majority of customers concentrated between 0 and 100 days, indicating that a significant portion of the customer base has purchased recently or within the last few months. The estimated median recency falls between 45 and 55 days, which suggests that while overall engagement is reasonable, a non-negligible share of customers is already showing signs of disengagement. The long tail extending beyond 300 days points to a smaller but relevant group of customers who have not interacted with NEW SMI for an extended period, representing a clear opportunity for targeted reactivation and win-back campaigns. This pattern reinforces the importance of recency as a key variable in both the RFM segmentation and the broader customer engagement strategy.

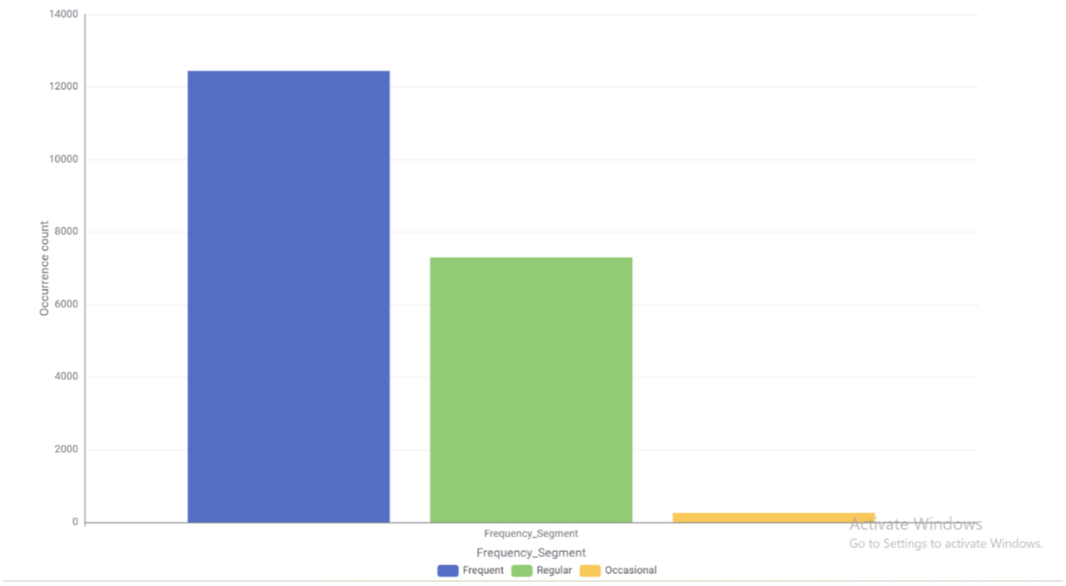

Figure 5 - Customer Distribution by Frequency Segment

The frequency segment distribution reveals that NEW SMI benefits from a predominantly active and repeat-buying customer base. Approximately 62% to 63% of customers fall into the frequent shopper category, purchasing more than 15 times per year, while around 36% to 37% are classified as regular shoppers, with between 5 and 15 annual purchases. Occasional shoppers, those purchasing fewer than 5 times per year, represent only 1% to 2% of the total base, a notably small share that reflects a generally engaged customer population. From a marketing standpoint, the priority lies in two directions: sustaining the loyalty of frequent shoppers through retention-focused initiatives and gradually converting regular shoppers into more committed buyers through personalized promotions and targeted engagement campaigns.

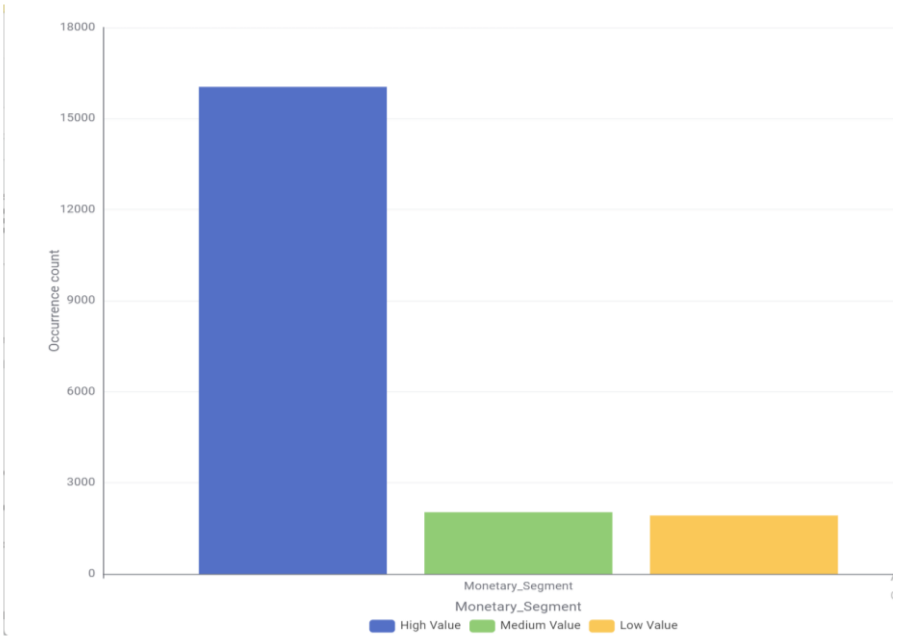

Figure 6 - Customer Distribution by Monetary Segment

The monetary segment distribution reveals that NEW SMI's customer base is predominantly composed of high-value spenders. Approximately 80% of customers are classified as High Value, spending more than €900 annually, while the Medium Value and Low Value segments each account for roughly 10% of the total base, corresponding to annual spending between €400 and €900, and below €400 respectively. This distribution indicates that the business is not reliant on low-spending customers but instead benefits from a large core of high-value buyers who likely account for the substantial majority of total revenue. From a strategic standpoint, this reinforces the importance of retention-focused initiatives aimed at protecting and rewarding this dominant segment, while simultaneously encouraging medium-value customers to increase their spending through personalized promotions, loyalty programs and targeted product recommendations.

1.2.2. Descriptive Statistics

The table below presents the main descriptive statistics for the numerical variables in the dataset, prior to the application of data quality corrections:

Table 2 - Descriptive Statistics table

Variable	Min	Max	Mean	Std. Dev.	Skewness	Kurtosis	NaN
Age	19	83	50.25	15.63	0.17	-1.17	0
Dependents	0	1	0.66	0.47	-0.68	-1.54	0
Income	0	191402	46401	18924	0.02	-0.48	35
Frequency	0	64	20.83	11.22	0.63	-0.38	0
Monetary	27	31939	4548	4223	1.02	0.22	0
Recency	0	373	61.17	58.69	3.21	12.84	0
Perishables	-2	13025	2119	2162	1.34	2.01	0
Beverages	-2	9299	830	1005	1.99	5.67	0
Frozen	-2	19656	580	808	3.42	16.58	0
Canned	-2	13074	499	798	4.05	33.59	0
Others	-2	13074	520	758	3.74	24.60	0
Internet	10	106	57.43	18.81	0.27	-0.96	0
NPS	1	5	3.43	1.02	-0.12	-1.07	0

1.2.3. Variable Correlation

The correlation analysis was performed using a Pearson correlation matrix across the main numerical variables in the dataset.

Table 3 - Pearson Correlation Matrix table

	Age	Income	Frequency	Monetary	Recency	Internet	NPS
Age	1	0.884	0.754	0.814	-0.168	0.785	0.460
Income	0.884	1	0.763	0.809	-0.175	0.700	0.465
Frequency	0.754	0.763	1	0.923	-0.191	0.594	0.569
Monetary	0.814	0.809	0.923	1	-0.152	0.723	0.578
Recency	-0.168	-0.175	-0.191	-0.152	1	-0.127	-0.091
Internet	0.785	0.700	0.594	0.723	-0.127	1	0.377
NPS	0.460	0.465	0.569	0.578	-0.091	0.377	1

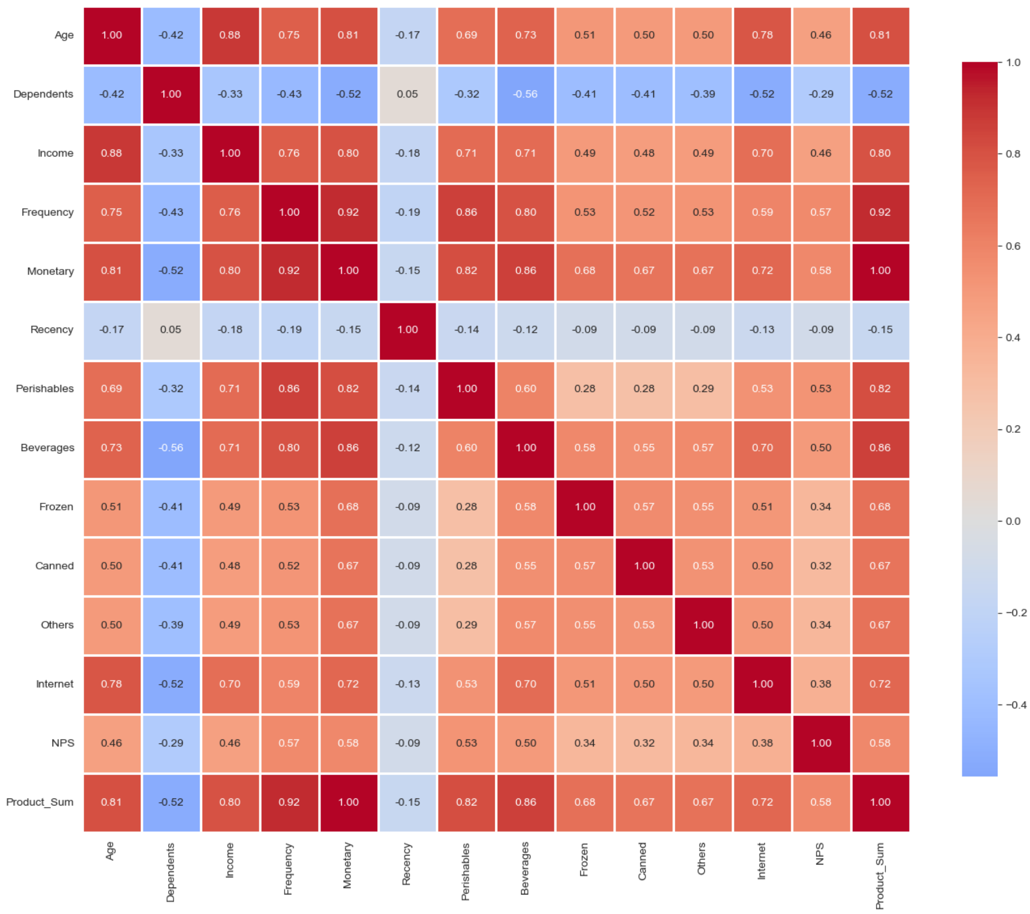

Figure 7 - Pearson Correlation Matrix figure

The correlation analysis reveals several noteworthy relationships across the dataset. The strongest association is observed between Frequency and Monetary (0.923), which is expected given that customers who purchase more often naturally tend to spend more overall. Age and Income also display a strong positive correlation (0.884), suggesting that older customers in this database tend to have higher household incomes, a pattern that is consistent with the Loyal High-Spenders profile identified in the segmentation stage.

Equally noteworthy is the relationship between Age and Internet (0.785), which challenges the common assumption that older customers are less digitally engaged and is further supported by the cluster analysis, where the oldest segment proved to be the most active online channel user.

Recency, by contrast, presents weak negative correlations across all other variables, which is coherent with its measurement logic: a higher recency value indicates more days since the last purchase, meaning it naturally moves in the opposite direction to engagement indicators such as Frequency and Monetary.

The Dependents variable shows consistent negative correlations across nearly all dimensions, most notably with Monetary (-0.52), Beverages (-0.56) and Internet (-0.52). This suggests that customers with dependents in the household tend to spend less, engage less through digital channels and purchase fewer beverages, a pattern that likely reflects tighter household budget constraints.

Among the product categories, Beverages stands out with the strongest associations with both Frequency (0.80) and Monetary (0.86), indicating that it is the category most closely linked to higher overall engagement and spending levels. Frozen, Canned and Others present moderate intercorrelations with each other, ranging between 0.53 and 0.57, suggesting that customers who spend on one of these categories tend to allocate spending across the others as well.

Finally, the pronounced correlations among Age, Income, Frequency, Monetary and Internet point to a degree of multicollinearity in the dataset, which informed the decision to apply a normalization step and a correlation filter prior to training the Neural Network models, ensuring that only non-redundant variables were included in the predictive modeling stage.

1.3. Data Preparation

1.3.1. Detect And Remove Outliers

In this step, we addressed extreme values that could distort our analysis, specifically within the income variable. Given that the income variable follows a normal distribution, the extreme outliers on the higher end were handled through a capping approach. Specifically, we substituted these outlier values with the maximum reasonable value within our expected range, which was set at 95,000€.

Additionally, we addressed anomalies in the product categories (perishables, beverages, frozen, canned, and others), where we detected 322 customers (1,61% of dataset) with negative values (0€ to -2€). We did a root cause analysis and found out that these negative numbers actually represented product returns and refunds from 2025. Because of how the original data was recorded, these returns showed up as negative transactions. To fix this, we decided to apply a floor function using the max(value, 0) rule. We implemented this in KNIME using the Column Expressions node, applying the function across all five product categories to automatically turn any negative values into zero. We confirmed that all product category values are now greater than or equal to zero, the total monetary value still perfectly matches the sum of all product categories for all 20K customers, and that our K-

Means distance calculations won't be distorted by negative numbers, giving us much more accurate clustering results.

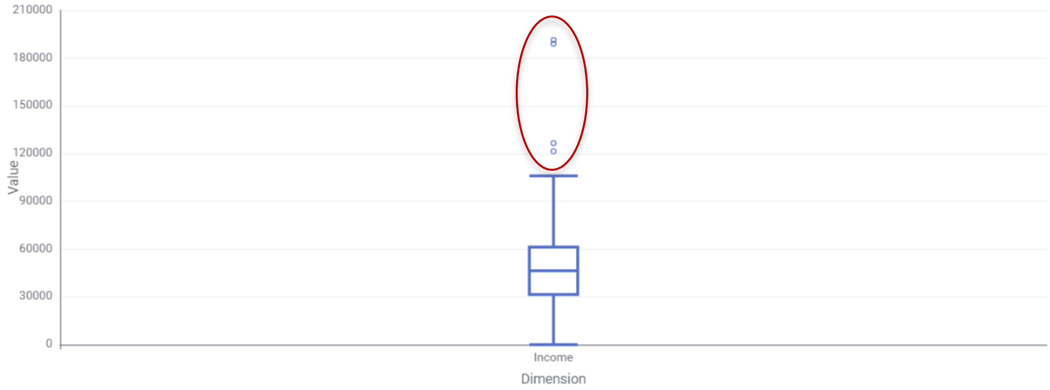

Figure 8 - Example of outliers on Variable Income

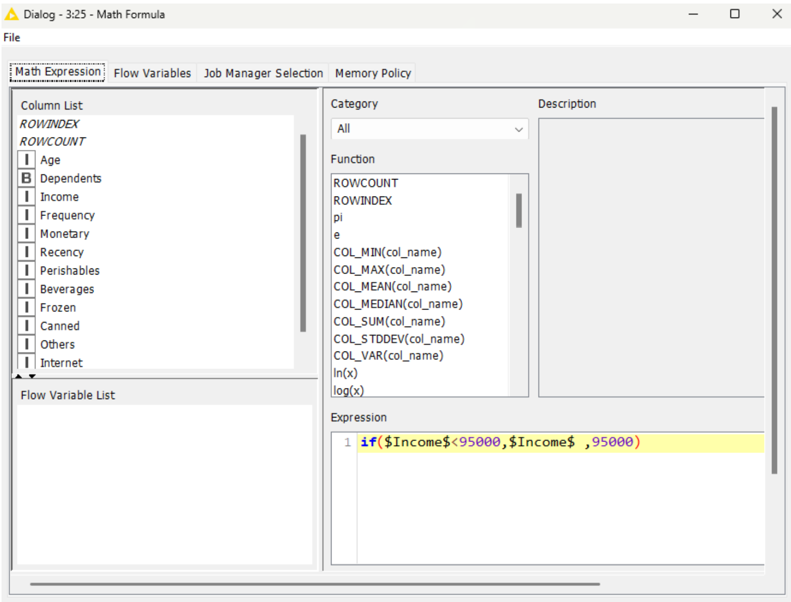

Figure 9 - Capping Method Applied to the Income Variable

For the Monetary variable, values were capped at €16,773, corresponding to the IQR upper bound, preventing extreme spending levels from dominating K-Means distance calculations. Regarding product categories, approximately 10% to 11% of customers showed outlier spending behavior, with 419 outliers detected in Perishables, 1,577 in Beverages, 2,106 in Frozen, 2,111 in Canned and 2,084 in Others. These values were not the result of data errors but rather a natural reflection of absolute spending levels, where high-value customers appeared as outliers across every category simply due to their overall spending power. To address this, percentage-based variables were created for each category, such as Perishables_Pct and Beverages_Pct, by dividing each category's spending by the customer's total monetary value. This approach captures true product preferences independently of spending power, and these percentage variables were used exclusively for clustering, while the original euro-denominated columns were retained for any future revenue analysis.

Finally, 322 customers (1.61% of the dataset) presented negative values across the product categories, ranging from 0€ to -2€. A root cause analysis determined that these figures represented product returns and refunds from 2025 that had been recorded as negative transactions due to the way the original data was structured. To address this, a floor function using the max(value, 0) rule was applied across all five product categories using the Column Expressions node in KNIME, replacing any negative value with zero. Following this correction, all product category values were confirmed to be greater than or equal to zero, and the total monetary value was verified to match the sum of all product categories across all 20K customers.

#	RowID	Column (String)	Min (Number (...	Max (Number (...	Mean (Number (...	Std. devi... (Number (...	Variance (Number (...	Skewness (Number (...	Kurtosis (Number (...
1	Age	Age	19	83	50.251	15.634	244.435	0.173	-1.17
2	Dependents	Dependents	0	1	0.661	0.473	0.224	-0.681	-1.536
3	Income	Income	0	191,402	46,401.353	18,924.533	358,137,942.13	0.023	-0.482
4	Frequency	Frequency	0	64	20.827	11.215	125.77	0.63	-0.378
5	Monetary	Monetary	27	31,939	4,548.197	4,223.234	17,835,709.004	1.016	0.215
6	Recency	Recency	0	373	61.174	58.687	3,444.178	3.209	12.839
7	Perishables	Perishables	-2	20,692	2,119.46	2,162.31	4,675,584.576	1.341	2.007
8	Beverages	Beverages	-2	13,025	829.572	1,004.791	1,009,605.032	1.989	5.666
9	Frozen	Frozen	-2	9,299	580.201	807.609	652,231.533	3.422	16.578
10	Canned	Canned	-2	19,656	499.312	798.441	637,507.889	4.05	33.585
11	Others	Others	-2	13,074	519.651	758.211	574,884.679	3.741	24.602
12	Internet	Internet	10	106	57.429	18.805	353.621	0.266	-0.96
13	NPS	NPS	1	5	3.429	1.02	1.041	-0.117	-1.065

Figure 10 - Identification of Negative Values in Product Categories

1.3.2. Replacements and Fill Missing Values

Data cleaning required addressing several missing values across multiple variables using specific strategic assumptions. For the income variable, we identified 35 missing values (0.18%. These were imputed using the overall mean income to maintain dataset completeness without introducing bias. Also 27 customers (0.14%) reported income = 0, indicating non-response rather than zero earnings. These were imputed using median income by Education level, as education is a predictor of income. This approach preserves realistic income distributions within each education segment (e.g., PhD median: €X, Bachelor median: €Y).

In the marital status variable, missing values were filled with "single," under the assumption that non-responses likely indicated this status. Similarly, all missing values for the gender variable were replaced with the category "other".

Name	Type	# Missing val...	# Unique val...	Minimum	Maximum	25% Quantile	50% Quantile...	75% Quantile
Education	String	0	6	?	?	?	?	?
Marital_Status	String	489	5	?	?	?	?	?
Age	Number (Integer)	0	65	19	83	36	49	64
Gender	String	169	3	?	?	?	?	?
Dependents	Boolean	0	1	0	0	0	0	0
Income	Number (Integer)	35	17406	0	191,402	31,495.5	46,431	61,300.5
Frequency	Number (Integer)	0	64	0	64	12	18	29
Monetary	Number (Integer)	0	9323	27	31,939	1,159	2,899.5	7,404.75
Recency	Number (Integer)	0	344	0	373	27	53	79
Perishables	Number (Integer)	0	6169	-2	20,692	350.25	1,387	3,302.75
Beverages	Number (Integer)	0	3507	-2	13,025	163	387	1,136
Frozen	Number (Integer)	0	2766	-2	9,299	154	268	658
Canned	Number (Integer)	0	2662	-2	19,656	67	185	571
Others	Number (Integer)	0	2650	-2	13,074	112	232	603
Internet	Number (Integer)	0	92	10	106	42	55	73
NPS	Number (Integer)	0	5	1	5	3	4	4

Figure 11 - Identification of Missing Values in Product Categories

For the educational variable, we found 222 missing values. These individuals were strategically assigned to the "Primary Education" category based on clear demographic trends. Since the majority of these non-respondents belong to an older cohort born before secondary education became compulsory, it is statistically probable that their formal education concluded at the primary level. This is also supported by the fact that people often avoid answering when they have lower education levels, due to social pressure.

1.4. Feature Engineering

1.4.1. Incoherence Check

To ensure data integrity, we performed incoherence checks between related variables. We initially created a new column to calculate the sum of all product variables and compared these results against the total monetary values recorded. After correcting the negative anomalies in the product categories, replacing them with 0, we re-ran this comparison, and the results remained consistent.

Additionally, we identified a system error where 10 rows showed a frequency of 0 (0.05%) but still recorded a positive monetary value (227€-13,232€). We discovered that the frequency variable was designed to represent the sum of in-store and internet orders, but the system had only recorded in-store orders. Consequently, customers who exclusively purchased online showed a frequency of 0 despite spending money.

To resolve this system incoherence, we set Frequency = 1 (minimum possible value) for these customers, because customers with positive spending must have made at least one purchase. Monetary values capped at 16,773€ (upper IQR bound) for K-Means clustering to prevent distance calculation domination and product categories will use percentage format to capture preferences independent of spending level.

1.4.2. Create New Variables and Variable Transformation

To improve the dataset's structure for modeling, we performed identifier substitution and variable transformations. First, we substituted the original RowID with CustID, as both variables carried the exact same meaning and transformed the dependents variable, converting it from a numerical value into a string boolean format to better represent the presence or absence of dependents (Figure 23).

We also created a new column, product_sum, which aggregates the total amount of money each customer spent across all individual product variables, allowing for a more comprehensive view of customer spending behavior.

To segment our audience, we implemented an RFM analysis by scoring each customer from 1 to 5 based on their Recency, Frequency, and Monetary values. By combining these metrics into a total score ranging from 3 to 15, we successfully divided our 20K customers into six strategic segments (Figure 24).

Our largest group is the At Risk segment, representing 32.5% of our database, which includes previously high-spending customers who haven't bought anything recently and are now our top priority for win-back campaigns. This is followed by the Others segment at 30.5%, which consists of mid-tier customers with average, mixed scores. On the positive side, we identified our highly engaged customers, split between Loyal customers at 14.5%, who show consistent high performance, and our

Champions at 9.25%, who are our absolute best VIP shoppers. Finally, the remaining data consists of Lost customers at 8%, who are inactive and historically low-value, and New customers at 5.25%, who recently made their first purchases and are perfect targets for onboarding offers.

Our biggest concern is that 32.5% of our customer base is currently at risk of churning, meaning we need to launch immediate retention campaigns to win them back. On the other hand, our top 23.75% of customers (Champions and Loyal segments) drive the majority of our revenue, so protecting them with exclusive loyalty programs is vital. Additionally, the 5.25% of new customers represent a fresh growth opportunity that requires targeted onboarding to increase their purchase frequency.

Based on these findings, we established three clear marketing priorities. Our number one focus is a massive win-back campaign targeting the 6,500 "At Risk" customers. Second, we will introduce a premium VIP program to reward and retain our 1850 "Champions." Finally, we will implement tailored welcome campaigns designed to build buying habits and increase frequency for our 1,050 "New" customers.

2. Customer Segmentation

The segmentation was built around eleven variables, grouped across three behavioral dimensions. Transactional behavior was captured through Frequency, Monetary, and Recency, which reflect how often, how much, and how recently each customer interacts with NEW SMI. Income and Age were added to provide sociodemographic context, helping to frame the purchasing patterns within a broader customer profile. Channel preference and category consumption were covered through Internet and the five product category ratios (Perishables_Pct, Beverages_Pct, Frozen_Pct, Canned_Pct, and Others_Pct).

Some variables, namely Gender; Education; Marital Status; and NPS, were left out of this stage. While Education, for instance, could reasonably relate to spending levels or purchase frequency, the decision was made to keep the segmentation grounded in direct behavioral and economic signals. That said, these variables are revisited in the profiling stage to add depth to each segment's interpretation.

2.1. Optimal Number of Clusters

To determine the optimal number of clusters, k-Means was run for K=2 through K=6, with the Mean Silhouette Coefficient (MSC) computed for each solution. Beyond the statistical output, cluster profiles were also reviewed to ensure the resulting segments were both interpretable and actionable from a business standpoint.

The results are presented in the table and figure below:

Table 4 - Mean Silhouette Coefficient by Number of Clusters table

Clusters	Mean Silhouette Coefficient
2	0,392
3	0,283
4	0,214
5	0,245
6	0,224

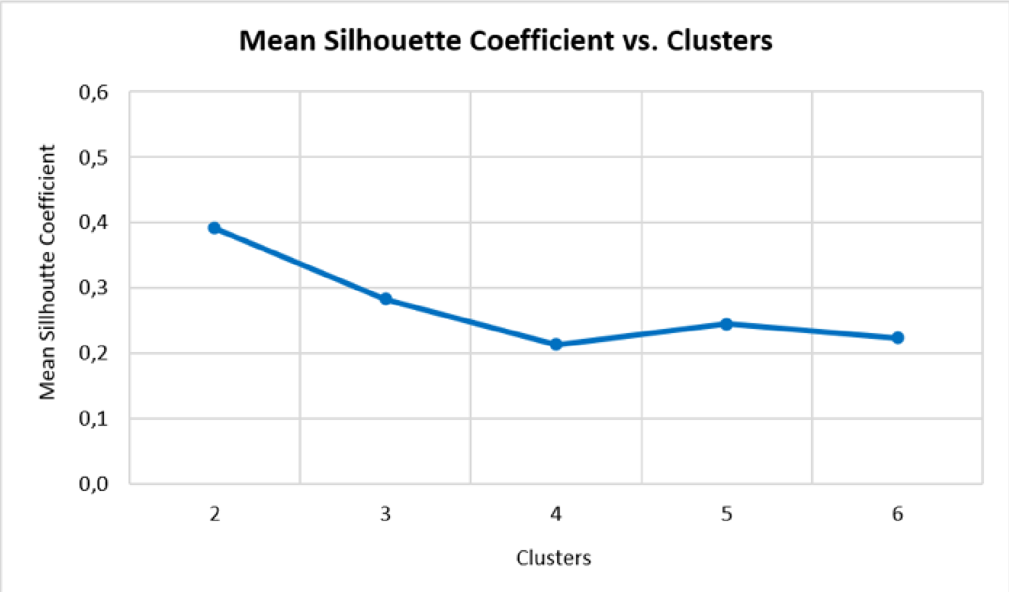

Figure 12 - Mean Silhouette Coefficient vs. Clusters line graph

While K=2 returned the highest MSC (0.392), a two-segment solution would offer little practical value, likely resulting in nothing more than a broad high versus low-value split, with limited room for differentiated marketing action.

K=3 was selected as the final solution. Despite a lower MSC (0.283), the three clusters produced distinct and meaningful profiles, with clear behavioral and sociodemographic differences that make them both interpretable and actionable. In line with the project's objectives, business relevance was weighed alongside statistical performance: segments need to make sense and drive decisions, not just score well on a metric.

2.2. Distance Between Clusters

To further validate the segmentation solution, the Euclidean distance between cluster centroids was computed using the Distance Matrix Calculate node in KNIME. The results are presented below:

Table 5 - Euclidean Distance Between Cluster Centroids table

Cluster Pair	Distance
Cluster 0 ↔ Cluster 1	40,483
Cluster 0 ↔ Cluster 2	18,859
Cluster 1 ↔ Cluster 2	21,839

All three pairs show meaningful separation, lending additional confidence to the chosen solution. The greatest distance is found between Cluster 0 and Cluster 1 (40,579), which points to strongly contrasting customer profiles between these two groups. Cluster 0 and Cluster 2 sit closest together (18,879), a degree of similarity that is worth noting and will be revisited when interpreting the segment profiles. **Across the board, the inter-cluster distances support the case for K=3 as a valid and well-differentiated segmentation.**

2.3. Cluster Analysis & Marketing Actions

2.3.1. Customer Profile

Following the selection of K=3 as the optimal segmentation solution, the three clusters were analyzed in terms of size and behavioral profile. Based on the analysis of each segment's behavioral and economic characteristics, the three groups were designated as Cluster 0 = Loyal High-Spenders, Cluster 1= Occasional Low-Spenders, and Cluster 2 = Regular Mid-Spenders. Their distribution across the 20K customers in the dataset is presented below.

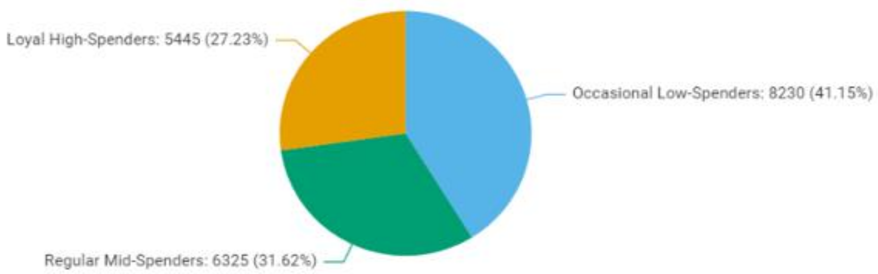

Figure 13 - Customer Distribution by Cluster pie chart

Table 6 - Customer Distribution by Cluster table

Cluster	Customers	
	Nº	%
Loyal High-Spenders	5 445	27.23%
Occasional Low-Spenders	8 230	41.15%
Regular Mid-Spenders	6 325	31.62%
Total	**20 000**	**100%**

No single segment dominates the customer base, with the largest group O*ccasional Low-Spenders* accounting for just over 40% of customers, and the smallest *Loyal High-Spenders* representing roughly one in four. This distribution is favorable from a marketing standpoint, as it allows for differentiated strategies without concentrating efforts on an overly narrow or overly broad audience.

The mean statistics per segment, presented in the charts below, reveal clear and consistent differences across all key dimensions.

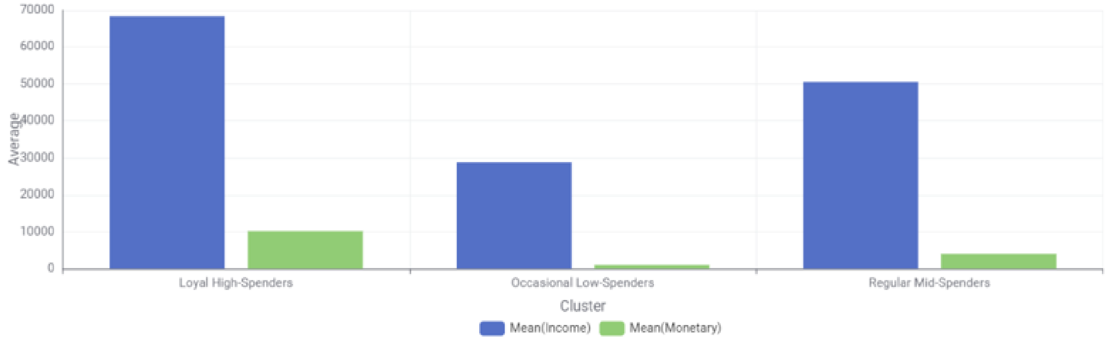

Figure 14 - Cluster profiles: Economic value grouped bar chart

From an economic standpoint, *Loyal High-Spenders* stand out with a mean annual income of €68,307 and mean spending of €10,279, figures that are substantially higher than the other two segments. Occasional *Low-Spenders* sit at the opposite end, with a mean income of €28,870 and annual spend of just €1,138. *Regular Mid-Spenders* occupy the middle ground, with a mean income of €50,518 and spending of €4,020.

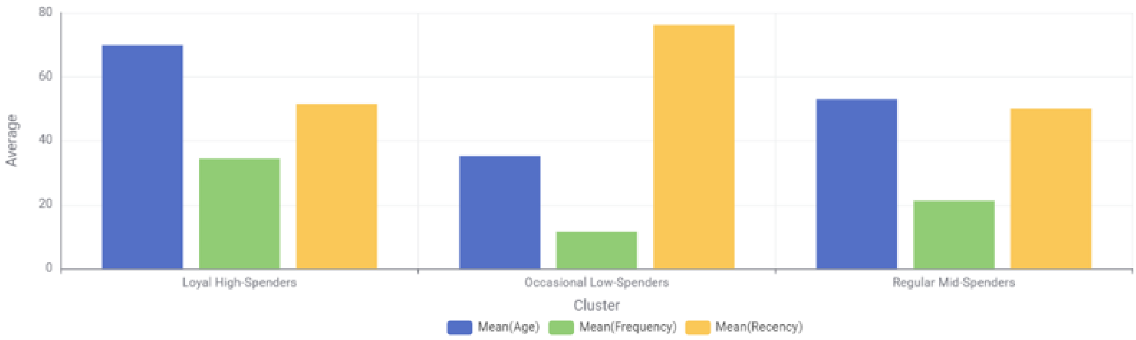

Figure 15 - Cluster Profiles: Purchase behavior grouped bar chart

Purchase behavior reinforces these differences. *Loyal High-Spenders* are the oldest group (mean age 70) and the most active, with an average of 34 purchases per year and a recency of 51 days. *Occasional Low-Spenders*, with a mean age of 35, purchase only 12 times per year on average and have the highest recency (76 days), indicating lower engagement with NEW SMI. *Regular Mid-Spenders* fall in between, averaging 21 purchases per year and a recency of 50 days.

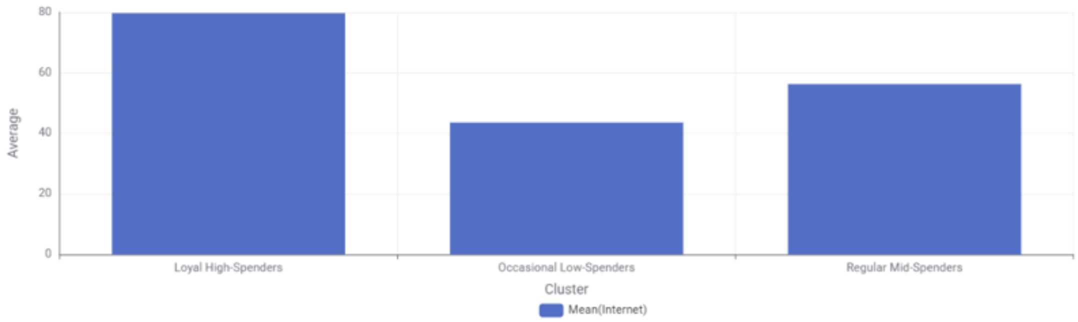

Figure 16 - Cluster Profiles: Online Channel grouped bar chart

In terms of channel preference, *Loyal High-Spenders* are the most digitally active segment, with nearly 80% of their purchases made online. *Regular Mid-Spenders* follow at 56%, while *Occasional Low-Spenders* show the lowest online engagement at 44%.

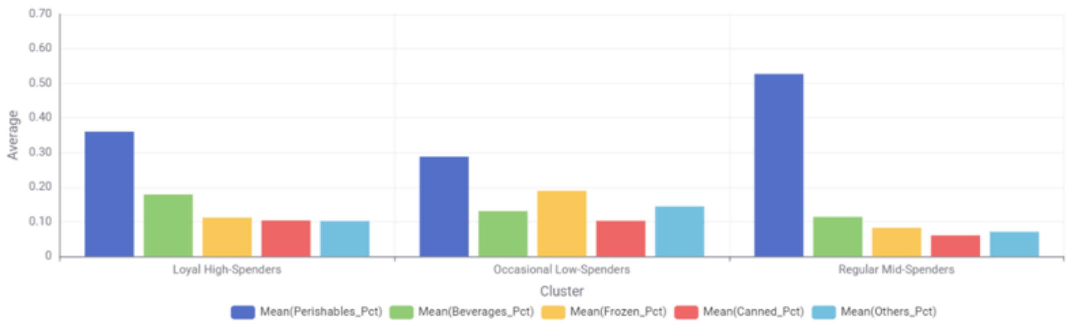

Figure 17 - Cluster Profiles: Category Mix grouped bar chart

Finally, spending patterns across product categories reveal a distinguishing trait for *Regular Mid-Spenders*: over 50% of their purchases fall within the Perishables category, the highest share across all three segments. *Loyal High-Spenders* also show a relatively high perishables share (36%), while *Occasional Low-Spenders* present a more balanced mix, with a slightly higher proportion of Frozen products compared to the other groups.

2.3.2. Profile Resume & Marketing Actions

LOYAL HIGH-SPENDERS

At 27.23% of the customer base, this is NEW SMI's most valuable segment. These are older customers (avg. 70 years old) with high household incomes (€68,307) who have built a consistent and frequent relationship with the brand: 34 purchases per year, an annual spend of €10,279, and a recency of just 51 days. What makes this segment particularly notable is their strong digital behavior with nearly 80% of their purchases online, which is remarkable given their age profile. Fresh produce dominates their basket (36%), followed by beverages (18%).

Marketing Actions:

Retention is the priority. NEW SMI should introduce a **premium loyalty program** offering exclusive benefits with personalized promotions, early access to deals, and dedicated service. Their online habits make them the natural first audience for a **delivery subscription service**, converting frequent one-off purchases into a recurring commitment. Communication should be digital first, through mobile app and email, with recommendations tailored to their fresh and beverage preferences.

OCCASIONAL LOW-SPENDERS

The largest segment at 41.15%, yet the least engaged. These are younger customers (avg. 35 years old) with lower incomes (€28,870) who shop infrequently (12 purchases per year) and have been away for an average of 76 days. Their annual spend of €1,138 leaves significant room for growth, and their relatively balanced category mix suggests no strong brand attachment to any product category. With 44% of purchases online, there is also untapped potential in the digital channel.

Marketing Actions:

The focus here is **reactivation and frequency**. Time-limited vouchers and discount campaigns targeting customers inactive for over 60 days can bring this segment back into the store or mobile app. Given their price sensitivity, **private label promotions and everyday essentials deals** are likely to resonate most. Social media and app push notifications are the most suitable channels to reach this younger, digitally aware audience.

REGULAR MID-SPENDERS

Representing 31.62% of customers, this segment is the backbone of NEW SMI's day-to-day business. With a mean age of 53 and a household income of €50,518, they shop 21 times per year and spend €4,020 annually, figures that sit comfortably in the middle of the three segments. What sets them apart is their strong attachment to fresh produce, which accounts for over 50% of their category spent: the highest share across all groups. They split their purchases evenly between in-store and online (56% digital).

Marketing Actions:

Their fresh produce habit is both a strength and an opportunity. A **recurring grocery subscription**: a weekly or bi-weekly fresh basket at a preferential rate would formalize and reward this behavior while locking in visits frequently. Beyond perishables, **cross-selling into frozen and canned categories** represents clear upside: bundled promotions pairing fresh items with discounts on underperforming categories can gradually broaden their basket. Given their mixed channel behavior, communication should span both in-store touchpoints and digital outreach.

3. Predictive Modeling

The second analytical component of this project addresses the challenge of identifying which customers are most likely to subscribe to NEW SMI's new delivery service. A binary classification framework was used, with the target variable Subs coded as 1 for likely subscribers and 0 otherwise. Two modelling approaches were explored, Decision Trees and Neural Networks, each put through multiple hyperparameter configurations in search of the best trade-off between fit and generalizability. Results were consistently evaluated on the validation set across five metrics: Accuracy, Recall, Precision, Cohen's Kappa and AUC.

3.1. Data Partition

The dataset was split into training (70%) and validation (30%) sets using stratified sampling, with Subs as the group column (Figure 25). Stratification ensures the proportion of subscribers (Subs=1) and non-subscribers (Subs=0) is preserved across both partitions, preventing class imbalance from skewing model performance during training and evaluation.

3.2. Decision Tree

Three Decision Tree models were tested with varying hyperparameters to identify the best configuration. All models used Reduced Error Pruning and 16 threads. The configurations and results are summarized below:

Table 7 - Decision Tree Models Performance Comparison

Model	Quality Measure	Min Records/Node	Accuracy	Recall	Precision	Kappa	AUC
DT 1	Gain Ratio	14	0.910	0.323	0.625	0.382	0.677
DT 2	**Gini Index**	**10**	**0.918**	**0.435**	**0.659**	**0.482**	**0.749**
DT 3	Gain Ratio	12	0.903	0.355	0.550	0.381	0.655

DT 2 stood out across all metrics, recording the highest Accuracy (0.918), Recall (0.435), Precision (0.659), Cohen's Kappa (0.482) and AUC (0.749). The Gini Index as the splitting criterion combined with a lower minimum records per node (10) allowed for finer splits, capturing more of the subscriber class without sacrificing overall accuracy.

3.3. Neural Networks

Two Neural Network models were tested using the RProp MLP algorithm. Given that Neural Networks are sensitive to variable scale, all inputs were normalised using Min-Max scaling prior to training. A Linear Correlation and Correlation Filter were applied to address multicollinearity, ensuring only non-redundant variables were fed into the models.

Table 8 - Neural Network Models Performance Comparison

Model	Hidden Layers	Neurons/Layer	Iterations	Accuracy	Recall	Precision	Kappa	AUC
NN 1	3	11	100	0.892	0.419	0.473	0.385	**0.851**
NN 2	3	11	150	0.893	0.435	0.482	0.399	0.833

NN 1 achieved the highest AUC (0.851) despite fewer iterations, suggesting that additional training in NN 2 did not translate into better discriminative power on the validation set. It is worth noting that an earlier configuration of NN 2 with 8 hidden layers and 5 neurons per layer produced an AUC of 0.882 but failed to identify any subscribers on the validation set, a clear sign of overfitting. The architecture was revised accordingly.

3.4. Model Selection - Comparison

Table 9 - Overall Model Performance Comparison: Decision Trees vs Neural Networks

Model	Accuracy	Recall	Precision	Kappa	AUC
DT 1	0.910	0.323	0.625	0.382	0.677
DT 2	0.918	0.435	0.659	0.482	0.749
DT 3	0.903	0.355	0.550	0.381	0.655
NN 1	0.892	0.419	0.473	0.385	**0.851**
NN 2	0.893	0.435	0.482	0.399	0.833

Neural Network 1 was selected as the final model, based on its superior AUC of 0.851. In a marketing campaign context, AUC is the most meaningful metric as it measures the model's ability to rank customers by their likelihood to subscribe, which directly determines the efficiency of targeted outreach. While DT 2 outperformed NN 1 in Accuracy and Precision, its substantially lower AUC (0.749) reflects weaker discriminative power overall.

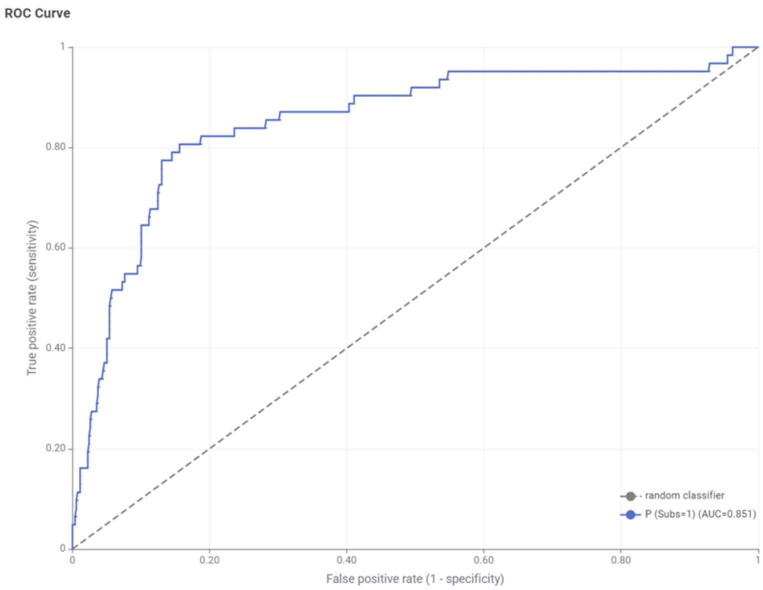

Figure 18 - ROC Curve Selected Model - NN 1, AUC 0.851

4. Profit Analysis

To assess the financial impact of the predictive model, a profit analysis was conducted comparing targeted outreach against a random contact strategy, applied to the validation set of 2,000 customers. Two assumptions were set by the team: a profit of €35 per successful subscription and a contact cost of €4.5 per customer reached, with a baseline response rate of 10.4%.

% Contacted	n Contacted	Cumulative lift	Cumulative Response Rate	Gross profit	Contact cost	Net profit
10%	200	4,80	49,9%	€ 3 494,40	€ 900,0	€ **2 594,40**
20%	400	3,80	39,5%	€ 5 532,80	€ 1 800,0	€ **3 732,80**
30%	600	2,70	28,1%	€ 5 896,80	€ 2 700,0	€ **3 196,80**
40%	800	2,20	22,9%	€ 6 406,40	€ 3 600,0	€ **2 806,40**
50%	1000	1,80	18,7%	€ 6 552,00	€ 4 500,0	€ **2 052,00**
60%	1200	1,60	16,6%	€ 6 988,80	€ 5 400,0	€ **1 588,80**
70%	1400	1,40	14,6%	€ 7 134,40	€ 6 300,0	€ **834,40**
80%	1600	1,30	13,5%	€ 7 571,20	€ 7 200,0	€ **371,20**
90%	1800	1,02	10,6%	€ 6 683,04	€ 8 100,0	-€ **1 416,96**
100%	2000	1,00	10,4%	€ 7 280,00	€ 9 000,0	-€ **1 720,00**

n		2000
baseline		10,4%
profit	€	35,0
cost	€	4,5

Figure 17 — Profit Cost Analysis

Targeting the top **20% of customers** (400) ranked by subscription propensity delivers the strongest return, with a net profit of approximately **€3,733** and nearly 40% of those contacted converting into subscribers. That is almost four times the 10.4% baseline response rate.

Moving beyond that threshold, each additional wave of customers contacted adds less value than it costs. The drop is gradual at first, with 30% still yielding €3,197, but accelerates from 80% onwards as the pool of likely subscribers runs dry. By the time the full customer base is reached, contact costs have swallowed the revenue entirely, leaving a net loss of **-€1,720**.

Comparing both extremes makes the case plainly: a targeted campaign reaching 400 customers generates €3,733, while a mass campaign reaching all 2,000 loses €1,720. The difference of nearly €5,500 represents the direct financial value of using the predictive model over contacting everyone indiscriminately.

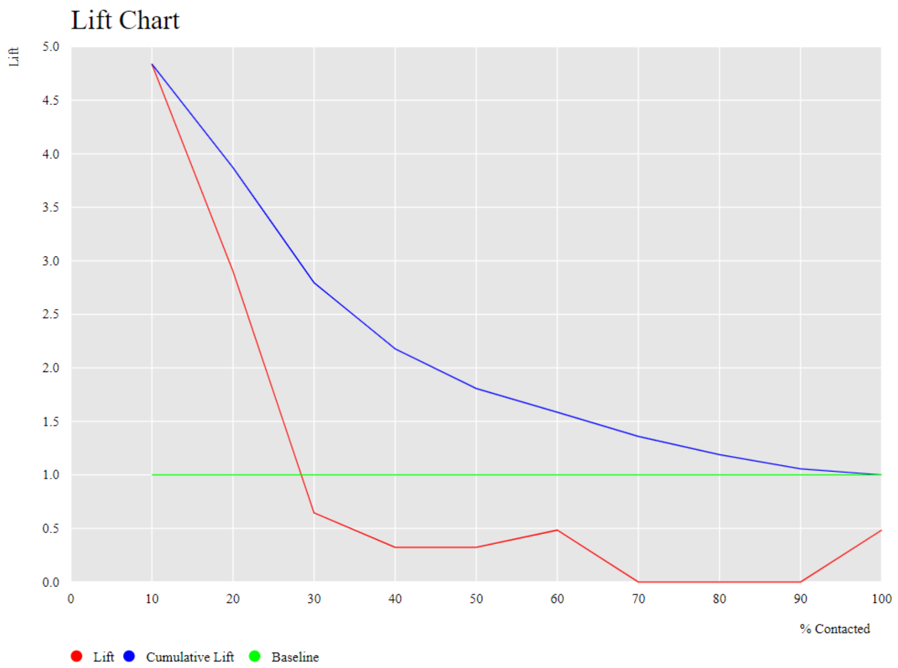

Figure 19 - Lift Chart

III. Conclusions

With over fifty years in the Portuguese retail market, NEW SMI has long operated without a structured understanding of who its customers truly are and what drives their purchasing decisions. The analyses conducted throughout this project change that, making clear that the company is not serving a single homogeneous audience but rather three structurally different customer groups, each with distinct economic profiles, behavioral patterns and channel preferences that demand fundamentally different marketing approaches.

Treating these groups as one is not only inefficient but actively costly. The profit analysis demonstrated this with a difference of nearly €5,500 between a targeted and an indiscriminate campaign, making the broader implication straightforward: mass marketing is no longer a neutral choice for NEW SMI, it is a financial liability.

Beyond the immediate campaign for the home delivery subscription service, the results of this project carry wider strategic significance. The segmentation framework gives the company a foundation it previously lacked, one that allows marketing investments to be allocated according to customer value rather than distributed evenly across a diverse base. Loyal High-Spenders, despite representing just over a quarter of the customer base, concentrate on the highest revenue potential and are the most natural audience for premium and digital-first initiatives. Occasional Low-Spenders, the largest group, require a different logic entirely, centered on reactivation and price sensitivity. Regular Mid-Spenders, with their strong attachment to fresh produce, offer clear cross-selling opportunities that a more targeted approach can begin to capture.

The predictive model further reinforces this shift, showing that the data NEW SMI already holds is sufficient to anticipate customer behavior with meaningful accuracy, provided it is used correctly. Perhaps most importantly, these findings suggest that the greatest untapped potential does not lie in acquiring new customers but in better serving and retaining the ones already in the database.

For these gains to be sustainable, the models developed here must be treated as living tools rather than one-off outputs, requiring regular retraining and monitoring as customer behavior evolves and new transactional data becomes available.

IV. Appendix

Column String	Min Number (...	Max Number (...	Mean Number (...	Std. devia... Number (...	Variance Number (...	Skewness Number (...	Kurtosis Number (...	Overall su... Number (...	No. missi... Number (I...	No. NaNs Number (I...
Age	19	83	50.251	15.634	244.435	0.173	-1.17	1,005,025	0	0
Dependents	0	1	0.661	0.473	0.224	-0.681	-1.536	13,224	0	0
Income	0	191,402	46,401.353	18,924.533	358,137,942.13	0.023	-0.482	926,403,015	35	0
Frequency	0	64	20.827	11.215	125.77	0.63	-0.378	416,531	0	0
Monetary	27	31,939	4,548.197	4,223.234	17,835,709.004	1.016	0.215	90,963,938	0	0
Recency	0	373	61.174	58.687	3,444.178	3.209	12.839	1,223,480	0	0
Perishables	-2	20,692	2,119.46	2,162.31	4,675,584.576	1.341	2.007	42,389,199	0	0
Beverages	-2	13,025	829.572	1,004.791	1,009,605.032	1.989	5.666	16,591,444	0	0
Frozen	-2	9,299	580.201	807.609	652,231.533	3.422	16.578	11,604,021	0	0
Canned	-2	19,656	499.312	798.441	637,507.889	4.05	33.585	9,986,250	0	0
Others	-2	13,074	519.651	758.211	574,884.679	3.741	24.602	10,393,024	0	0
Internet	10	106	57.429	18.805	353.621	0.266	-0.96	1,148,578	0	0
NPS	1	5	3.429	1.02	1.041	-0.117	-1.065	68,585	0	0

Figure 20 - Descriptive Statistics table

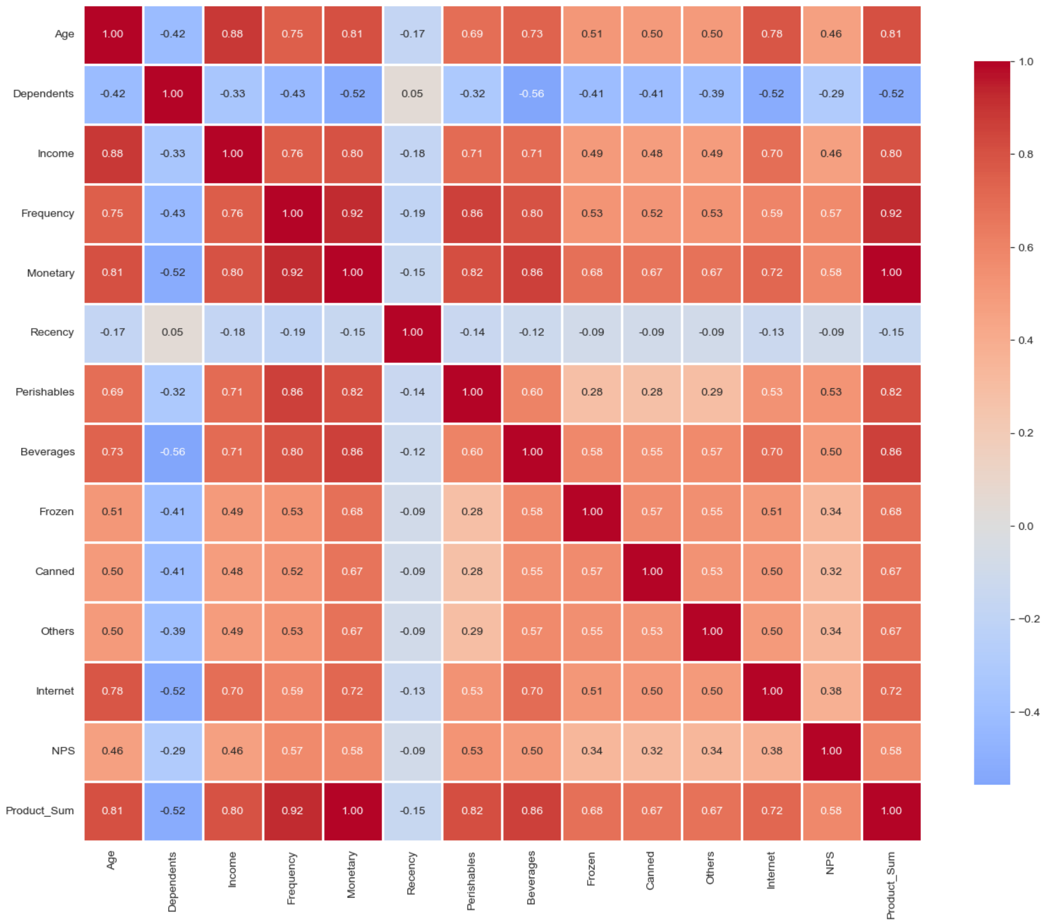

Figure 21 - Pearson Correlation Matrix figure

Correlation matrix (Table)

Rows: 7 | Columns: 7

#	RowID	Age Number (Float)	Income Number (Float)	Frequency Number (Float)	Monetary Number (Float)	Recency Number (Float)	Internet Number (Float)	NPS Number
1	Age	1	0.884	0.754	0.814	-0.168	0.785	0.46
2	Income	0.884	1	0.763	0.809	-0.175	0.7	0.465
3	Frequency	0.754	0.763	1	0.923	-0.191	0.594	0.569
4	Monetary	0.814	0.809	0.923	1	-0.152	0.723	0.578
5	Recency	-0.168	-0.175	-0.191	-0.152	1	-0.127	-0.091
6	Internet	0.785	0.7	0.594	0.723	-0.127	1	0.377
7	NPS	0.46	0.465	0.569	0.578	-0.091	0.377	1

Figure 22 - Pearson Correlation Matrix Table

CustID		Number (Integer)
Education		String
Marital_Status		String
Age		Number (Integer)
Gender		String
Dependents		Number (Integer)
Income		Number (Integer)
Frequency		Number (Integer)
Monetary		Number (Integer)
Recency		Number (Integer)
Perishables		Number (Integer)
Beverages		Number (Integer)
Frozen		Number (Integer)
Canned		Number (Integer)
Others		Number (Integer)
Internet		Number (Integer)

Figure 23 - Dependents Variable Type Transformation: Number (Integer) to String Boolean

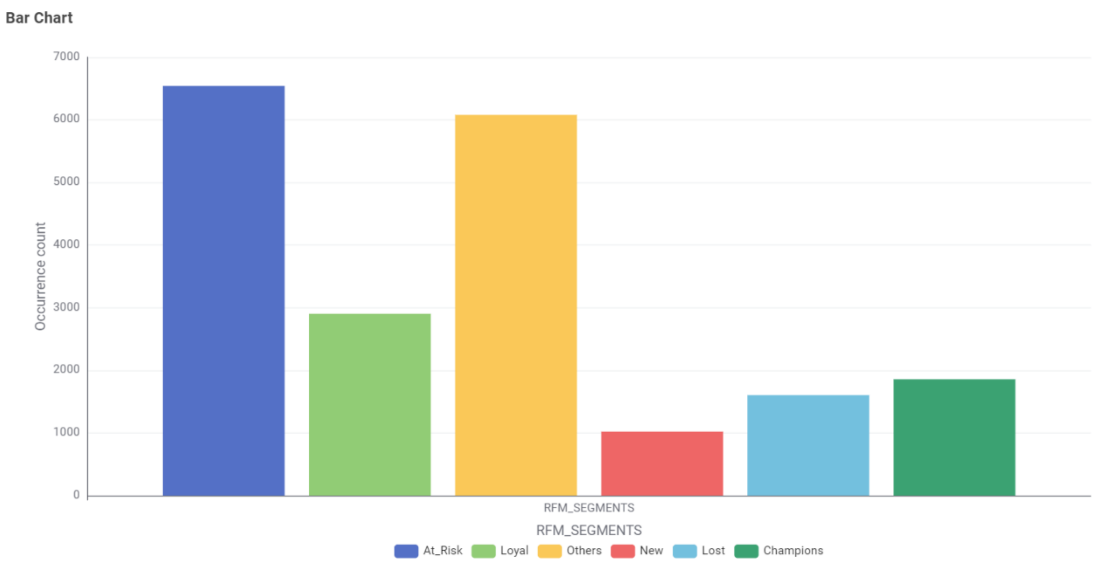

Figure 24 - Customer Distribution by RFM Segment Following Manual Segment Labelling

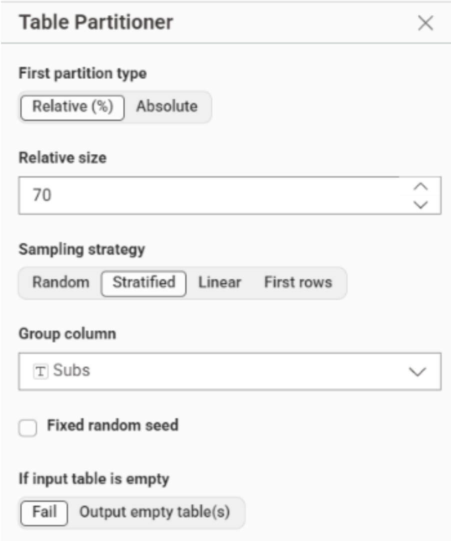

Figure 25 - Data Partition: Stratified Sampling Distribution (70% Training / 30% Validation)

Rows: 2 | Columns: 2

#	RowID	1 Number (Integer)	0 Number (Integer)
1	1	20	42
2	0	12	526

Rows: 3 | Columns: 11

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegatives Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	1	20	12	526	42	0.323	0.625	0.323	0.978	0.426	?	?
2	0	526	42	20	12	0.978	0.926	0.978	0.323	0.951	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.91	0.382

Figure 26 - Decision Tree 1: Confusion Matrix and Performance Metrics

Rows: 2 | Columns: 2

#	RowID	1 Number (Integer)	0 Number (Integer)
1	1	27	35
2	0	14	524

Rows: 3 | Columns: 11

#	RowID	TruePositives Number (Integ...	FalsePositives Number (Integ...	TrueNegatives Number (Integ...	FalseNegatives Number (Integ...	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	1	27	14	524	35	0.435	0.659	0.435	0.974	0.524	?	?
2	0	524	35	27	14	0.974	0.937	0.974	0.435	0.955	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.918	0.482

Figure 27 - Decision Tree 2: Confusion Matrix and Performance Metrics

Rows: 2 | Columns: 2

#	RowID	1 Number (Integer)	0 Number (Integer)
1	1	22	40
2	0	18	520

Rows: 3 | Columns: 11

#	RowID	TruePositives Number (Integ...	FalsePositives Number (Integ...	TrueNegatives Number (Integ...	FalseNegatives Number (Integ...	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	1	22	18	520	40	0.355	0.55	0.355	0.967	0.431	?	?
2	0	520	40	22	18	0.967	0.929	0.967	0.355	0.947	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.903	0.381

Figure 28 - Decision Tree 3: Confusion Matrix and Performance Metrics

Rows: 2 | Columns: 2

#	RowID	1 Number (Integer)	0 Number (Integer)
1	1	26	36
2	0	29	509

Rows: 3 | Columns: 11

#	RowID	TruePositives Number (Integ...	FalsePositives Number (Integ...	TrueNegatives Number (Integ...	FalseNegatives Number (Integ...	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	1	26	29	509	36	0.419	0.473	0.419	0.946	0.444	?	?
2	0	509	36	26	29	0.946	0.934	0.946	0.419	0.94	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.892	0.385

Figure 29 - Neural Network 1: Confusion Matrix and Performance Metrics

Rows: 2 | Columns: 2

#	RowID	1 Number (Integer)	0 Number (Integer)
1	1	27	35
2	0	29	509

Rows: 3 | Columns: 11

#	RowID	TruePositives Number (Integ...	FalsePositives Number (Integ...	TrueNegatives Number (Integ...	FalseNegatives Number (Integ...	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	1	27	29	509	35	0.435	0.482	0.435	0.946	0.458	?	?
2	0	509	35	27	29	0.946	0.936	0.946	0.435	0.941	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.893	0.399

Figure 30 - Neural Network 1: Confusion Matrix and Performance Metrics